

Ahmed FAKHFAKH

Junior Data & AI Engineer seeking a Permanent Position or CIFRE PhD starting October 1st, 2026

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Certifications

Google Cloud Platform (GCP): Professional Data Engineer | Associate Cloud Engineer | Cloud Digital Leader

Work Experience

- Oct. 2023 – Sep. 2026 **Edge AI Engineer Apprentice, OPmobility, Levallois-Perret**
- **R&D Deep Learning:** Design and AI-based processing of 4D radar data for Advanced Driver Assistance Systems.
 - **Architecture & Compression:** Redesigned a heavy network into a lightweight Sparse architecture.
 - **Knowledge Distillation:** Trained a Sparse model by imitating a Dense model to maintain detection accuracy on weak signals.
 - **Edge AI & Real-Time Deployment:** Performed INT8 quantization and compiled models via NVIDIA TensorRT on a Jetson target, achieving latency objectives.
 - **Software Engineering & Multiprocessing:** Implemented a FIFO pipeline that eliminated redundant computations and improved algorithm performance.
 - **Tracking & Reliability:** Integrated a Kalman filter in post-inference to smooth 3D trajectories and compensate for AI detection gaps.
 - **System Benchmarking & Profiling:** Conducted an evaluation of algorithmic performance coupled with hardware analysis.
- Technical Stack:** Python, PyTorch, TensorRT, NVIDIA Jetson, Edge AI, Multiprocessing, Kalman Filter.

Advanced Technical Projects

- 2025 – 2026 **Multimodal Financial Agentic AI (Multi-Agent Architecture & MLOps)**
- Designed a multi-agent system (LangGraph) dynamically orchestrating Big Data queries (BigQuery with self-correction), financial document RAG (ChromaDB), and real-time APIs.
 - Security & Observability: Implemented an AI firewall (Guardrails against prompt injection) and real-time cost/latency tracing in production (LangSmith).
 - Serverless Deployment & IaC: Containerized (Docker) on Google Cloud Run with asynchronous streaming (Streamlit) and infrastructure managed via Terraform.
- Stack & Tools:** GCP (Cloud Run, BigQuery), Vertex AI (Gemini), LangChain / LangGraph, ChromaDB, LangSmith, Terraform, Streamlit.
- 2025 – 2026 **Modern Data Stack & Advanced RAG (Mathematical Evaluation)**
- Designed a comprehensive ELT data engineering pipeline orchestrated by Apache Airflow, including data cleaning and transformation using dbt.
 - Developed a production-ready Retrieval-Augmented Generation (RAG) system with LlamaIndex, integrating a scalable vector database (Qdrant).
 - Conducted rigorous mathematical evaluation of RAG performance and hallucination detection using the RAGAS framework.
- Stack & Tools:** Apache Airflow, dbt, LlamaIndex, Qdrant, RAGAS, Python.

Education

- Oct. 2023 - Sep. 2026 **Data & AI Engineering Degree, Polytech Nantes, France**
Courses: Data Science, AI, Deep Learning, Image Processing, MLOps.
- Sep. 2021 - Jun. 2023 **Two-year Technical Degree in Computer Science, IUT de Bayonne et du Pays Basque, France**
Courses: Mathematical Analysis, Algebra, Algorithms, OOP, SQL, Competitive Programming.

Skills

Languages & Cloud: Python, SQL, Bash, R, GCP (BigQuery, Vertex AI, Cloud Run).

AI & GenAI: PyTorch, Scikit-Learn, XGBoost, LangGraph, LlamaIndex, Qdrant, HuggingFace (LoRA).

Data Eng & MLOps: Apache Airflow, dbt, MLflow, DVC, LangSmith, RAGAS.

DevOps & Tools: Terraform, Docker, CI/CD (Gitlab/GitHub Actions), FastAPI, Git.

Soft Skills: Solutions Architecture, Complex Problem Solving, Analytical Mindset.

Languages

English (TOEIC: 830) | French | Arabic